Wanchen Lian

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Education

Hong Kong University of Science and Technology (Guangzhou) Sep 2024 – Jan 2027
MPhil candidate, Artificial Intelligence

Selected coursework: AI Security and Privacy; Reinforcement Learning and Optimization; Frontier of "AI for Healthcare."

Beijing Normal University Sep 2020 – Jun 2024
Bachelor's degree, Chinese Language and Literature

Research Focus

Reliable AI for health: structured prediction and uncertainty estimation; evidence-grounded agents; human review, data governance, and evaluation of AI-enabled healthcare workflows.

Selected Publications and Research

Strengthening LLMs for Tabular Prediction with Structural Priors [Paper]
Pengxiang Cai, Zihao Gao, **Wanchen Lian**, Guocong Li, Jintai Chen.
NeurIPS 2026, Main Track Poster — accepted. Co-first author.

- Coauthored research on structural-prior LLM post-training. PRPO combines label-preserving column permutations with intra-/inter-permutation advantage estimation to densify sparse rewards and stabilize policy optimization.
- Across 139 OpenML datasets, the study's 8B model reaches **84.36%** mean accuracy on 50 fully supervised classification tasks versus 82.34% for XGBoost (**+2.02 percentage points**); cross-dataset transfer reaches 70.21% on 53 unseen classification tasks without target-dataset examples.

PATH: Next-Interval Prediction via Autoregressive Tree Hierarchy on Tabular Data [Paper]
Pengxiang Cai*, **Wanchen Lian***, Chenyang Liu, Xiaohan Li, Qingyuan Zeng, Jinhong Wang, Jintai Chen.
*Preprint; under review at AAAI 2027. *Equal contribution.*

- Contributed to architecture design and iterative refinement of PATH: **FT-Transformer** feature encoding, autoregressive residual corrections to tree-branch probabilities, and shortest-mass interval extraction.
- The architecture combines leaf, prefix, and cumulative-distribution losses with validation-based temperature, mass, and residual-scale selection. In the study's modeling ablation, mean normalized interval length falls from **0.1623** for a direct Transformer to **0.1473** for PATH (**9.2% lower**).
- Across **56 datasets, 10 seeds, and 24 baselines**, PATH's mean normalized interval length is **27.0% lower** than QRF-CTI (0.1473 vs. 0.2017), with **91.44%** empirical coverage at a nominal 90% target.

HipGo: Multimodal Smart Agent Solution for Remote Hip Dysplasia Care [Paper]
BIBM Workshop; first-author manuscript. Medical imaging and decision-support prototype.

- Implemented a **ResNet-50/FPN + nine-node GAT** for pelvic landmark localization, integrating image features and anatomical relationships with CE/Sharp/Tönnis angle measurements and multimodal report generation.
- The 800-X-ray study reports **PCK@0.1 of 0.86 vs. 0.72** for ResNet-50 (**+14 percentage points**) and **mAP 0.91**. Related agent development integrates case parsing, evidence retrieval, reports, and rehabilitation.

TBC-DDH: Gait-based Inverse Inference for Adult DDH Preoperative Diagnosis [Paper]
BIBM Workshop; first-author manuscript. Retrospective proof of concept.

- Investigated inverse inference from postoperative gait using multimodal spatial features, temporal rehabilitation modeling, and reverse-sequence embeddings. The **29-patient** study reports **86.67% ± 6.67%** accuracy; removing reverse-sequence embeddings reduces accuracy by 13.33 percentage points.

AI Systems Engineering and Applied Data Experience

Orient Feeling, Hangzhou [Details] Sep 2026 – Present

AI Agent Engineer — sleep-health agents and connected devices

- Designed task-scoped context and specialist-tool orchestration across intent routing, health data, specialist agents, and aggregation; enforced **user/subject permissions** and separated model analysis from confirmation-gated execution.
- Established trace-based RAG diagnostics across **26 product scenarios**, decomposing failures by retrieval, reranking, evidence delivery, and generation to distinguish source-coverage gaps from unsupported answer generation.
- Diagnosed wearable-to-agent score failures, implementing **source-aware JSON extraction with explicit missing-value semantics**; reconciled stored values against provider readbacks and contract regression checks.
- Modeled IoT control as **capability discovery, authorization, confirmation, and execution-state feedback**; compared direct API and agent-driven paths and checked single-action control for three device types in QA.

Baidu AI Cloud, Beijing [Details] Jul – Sep 2026

Medical Industry Strategy FDE — data modeling and multi-source intelligence

- Modeled pharma, medtech, and biotech organizations as **linked entities, R&D pipelines, procurement events, and sales states**; defined identity-resolution, deduplication, relationship, and history-traceability rules.
- Designed multi-agent collection and incremental-refresh workflows with **evidence verification and ingestion quality gates**; linked CRM, procurement, and R&D signals into a consolidated account view for medical-sector planning.

Qianshu Wenyi Technology, Guangzhou [Details] May – Jul 2026

Medical AI FDE Engineer — medical-record quality control

- Converted **10+ hospital-stakeholder interviews** into a QC workflow and data model linking defects, rule evidence, scoring, clinician responsibility, rectification deadlines, and review history.
- Implemented a runnable **JavaScript/HTML/CSS** prototype spanning AI screening, human adjudication, rectification, and score recalculation; tied review decisions to evidence, grading, reports, and operation logs.

Canchen Technology, Shenzhen [Details] Jan – May 2026

Multi-Agent Product Developer — rare-disease and scientific research workflows

- Designed tool routing and structured state for rare-disease analysis, carrying case summaries, **HPO phenotypes, candidate diseases, evidence bundles, and audit notes** across HPO, OMIM, Orphanet, and PubMed.
- Defined evaluation checks for phenotype extraction, candidate recall, citation quality, and tool execution; contributed **PaperAgent/GraphAgent** workflows and a research workspace separating reasoning, evidence, results, and files.

HiDream.ai, Shanghai [Details] Jun – Sep 2024

AI E-commerce Product Intern — multimodal generation and evaluation

- Built an image/video **failure taxonomy** for subject consistency, text alignment, lighting, composition, and temporal coherence; curated training and evaluation data through deduplication, denoising, and quality filtering.
- Iterated prompt templates, **classifier-free guidance (CFG)**, negative constraints, and localization-aware style controls; converted recurring failure modes into repainting, style-transfer, and image-cleanup requirements.

ByteDance, Shanghai [Details] Dec 2023 – Jun 2024

Strategy Product Intern / Data Analyst — e-commerce operations

- Decomposed merchant profitability, SKU mix, inventory depth, fulfillment reliability, and advertising ROI into comparable operating dimensions to guide assortment and allocation across **38 merchants**.
- Built **Aeolus / Power BI** dashboards for inventory and price-risk monitoring; classified after-sales feedback into recurring product issues and fed findings into SKU operations, stock actions, and campaign adjustments.

Technical Toolkit

Systems: Python, SQL, FastAPI, LangGraph, PostgreSQL/pgvector, JavaScript, HTML/CSS.
Methods / applications: RL, tabular modeling, CNN–GAT, RAG evaluation; medical QC, health agents, and IoT.